Kompetenzen

EXCEL

TABLEAU & POWER BI

SQL

PYTHON

MACHINE LEARNING

CAREER

KOMPETENZEN TECHNOLOGIES

Data Analytics with AI

125-Day Project-Driven Career Program

250 Technical Hours · 7 Phases

125 DAYS **250** TECHNICAL HOURS **04** MAIN PROJECTS

> Clean. Analyse. Visualise. Predict.

Kompetenzen Technologies

PROGRAMME OVERVIEW

Programme at a glance

A project-driven analytics programme that takes you from Excel fundamentals to dashboards, SQL, Python and machine learning — with AI built into how you analyse.

125	250	2	Project-driven	04
DAYS	TECHNICAL HOURS	HOURS PER DAY	LEARNING MODEL	MAIN PROJECTS

WHAT THIS PROGRAMME SETS OUT TO DO

01 Learn to clean and analyse data.

02 Learn to build dashboards.

03 Learn SQL and Python for data.

04 Learn machine learning and AI for analytics.

05 Prepare for the data analyst job market.

PREREQUISITE

Basic computer skills and comfort with numbers. The curriculum starts from Excel fundamentals.

HOW YOU LEARN

Learn, practise, build and review — every module ends in practical work.

THE IDEAL LEARNER

Who is this programme for?

This programme is suitable for students who:

- Want to become professional data analysts
- Enjoy working with numbers, patterns and business questions
- Are ready to learn from Excel fundamentals upward
- Want hands-on skills in Excel, Tableau, Power BI, SQL and Python
- Want to understand machine learning and AI for analytics
- Want real-world dataset projects and interview preparation

PLEASE NOTE

You do not need prior professional experience. You do need commitment: 2 hours of focused training every day for 125 days, plus practice.

THE LEARNING LOGIC

Why Data Analytics with AI?

01 EXCEL FOUNDATION

02 TABLEAU DASHBOARDS

03 POWER BI REPORTING

04 SQL ANALYSIS

05 PYTHON & EDA

06 MACHINE LEARNING

07 AI FOR ANALYTICS

Every business decision runs on data. This programme takes you from spreadsheets to dashboards, SQL and Python — and then to machine learning and AI, so you can move from reporting what happened to predicting what happens next.

The goal is not simply to use AI tools.
Understand the data. Use AI to analyse it faster.

WHAT YOU WILL WORK WITH

Technology stack

SPREADSHEETS

Excel · Pivot Tables · Power Query · Power Pivot · Macros & VBA basics · Solver

BUSINESS INTELLIGENCE

Tableau · Power BI · Power Query ETL · DAX · Row-level security

DATABASE

SQL · MySQL

PYTHON FOR DATA

Python · NumPy · Pandas · Matplotlib · Seaborn · Regular expressions

MACHINE LEARNING & AI

scikit-learn · KNN · Naïve Bayes · SVM · Linear & Logistic Regression · Decision Trees · Random Forest · XGBoost · K-Means · PCA · Generative AI for analytics · Prompt engineering for SQL & Python

GOOD TO KNOW

Three BI tools, one analyst mindset.

Students learn **Excel, Tableau and Power BI** — and combine all three in one real-world dashboard project.

SQL and Python then take them from reporting to analysis and machine learning.

7 PHASES · 125 DAYS

Your programme journey



Day 1 | Day 46 · Main Project 1 | Day 125 · Program end

01	EXCEL FOR DATA ANALYSIS	Days 1–14	28 Hrs
02	TABLEAU	Days 15–25	22 Hrs
03	POWER BI & MULTI-TOOL DASHBOARD	Days 26–49	48 Hrs
04	SQL FOR DATA ANALYSIS	Days 50–70	42 Hrs
05	PYTHON PROGRAMMING	Days 71–88	36 Hrs
06	PYTHON LIBRARIES & EDA	Days 89–103	30 Hrs
07	AI, MACHINE LEARNING & GENAI	Days 104–125	44 Hrs

TOTAL **125 Days** **250 Technical Hours**

Highlighted phases end with a major project.

PHASE 01

01 Excel for Data Analysis

Days 1–14 | **28** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07

Day 1 — Day 46 · Main Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will learn to clean, analyse and summarise data in Excel — from formulas and lookups to pivots, Power Query and basic automation.

KEY TOPICS

- Interface
- Data types
- Data entry
- Formatting
- Sorting
- Filtering
- SUM, AVERAGE, COUNT
- MIN, MAX
- LEFT, RIGHT, MID
- TRIM, CONCAT
- Date & time functions
- IF, IFS
- COUNTIF, SUMIF
- VLOOKUP, HLOOKUP
- XLOOKUP
- INDEX + MATCH
- Missing data
- Formula auditing
- Error handling
- Pivot Tables & Charts
- What-If Analysis
- Solver
- Forecasting
- Macros & VBA basics
- Power Query import & transform
- Power Pivot
- Data validation
- Remove duplicates
- Text to columns
- Guided project
- End-to-end analysis

PRACTICAL OUTCOMES

01 Formula & lookup workbooks

02 Pivot analysis

03 Power Query data model

04 Excel mini project

PHASE 02

02 Tableau

Days 15–25 | **22** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07

Day 1 | Day 46 · Main Project 1 | Day 125 · Program end

WHAT YOU WILL LEARN

You will connect data in Tableau, build calculations and LOD expressions, and publish interactive dashboards that tell a story.

KEY TOPICS

- Interface & data connection
- Dimensions vs measures
- Discrete vs continuous
- Calculated fields
- Parameters, sets & bins
- Aggregations
- Tableau architecture
- LOD expressions
- Dashboards
- Chart toggling
- Storytelling
- Publishing
- Guided project
- End-to-end dashboard

PRACTICAL OUTCOMES

01 Interactive Tableau dashboard

02 Data story

03 Tableau mini project

PHASE 03

03 Power BI & Multi-Tool Dashboard

Days 26–49 | **48** HOURS

01 | 02 | **03** | 04 | 05 | 06 | 07

Day 1 — Day 46 · Main Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will learn visual storytelling, Power Query ETL, data modelling and interactive Power BI reports — and build one dashboard across all three tools.

KEY TOPICS

Visualisation fundamentals · Chart selection · Storytelling · Power BI interface & workflow · Power Query ETL · Group by · Duplicate & reference queries · Data modelling · Relationships · DAX measures · CALCULATE & FILTER · Time intelligence · DAX functions · Filters & slicers · Drill-down · Visualisations · Row-level security · Demo project · Power BI mini project · Excel · Power BI · Tableau · Real-world dataset · Mock interview

PRACTICAL OUTCOMES

01 Power BI ETL pipeline

02 Data model

03 Secured interactive report

04 Multi-tool real-world dashboard

PHASE 04

04 SQL for Data Analysis

Days 50–70 | **42** HOURS

01 | 02 | 03 | **04** | 05 | 06 | 07

Day 1 — Day 46 · Main Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will write SQL from basic queries to joins, subqueries, CTEs and procedures, and connect a database to Power BI.

KEY TOPICS

Databases & relational concepts · Data integrity · Normalisation · SELECT, WHERE, DISTINCT · ORDER BY · GROUP BY, HAVING · AND, OR, NOT, LIKE, IN, BETWEEN · Constraints · Multi-table joins · Aggregate functions · Date & null functions · String functions · Subqueries · CASE · Views · CTEs (WITH) · Stored procedures basics · Connecting external data · Analysis · SQL to Power BI Desktop · Dashboard · SQL · Power BI · Real-world data

PRACTICAL OUTCOMES

- **01** SQL query sets
- **02** Multi-table analysis
- **03** SQL-powered Power BI dashboard
- **04** SQL main project

PHASE 05

05 Python Programming

Days 71–88 | **36** HOURS

01 | 02 | 03 | 04 | **05** | 06 | 07

Day 1 | Day 46 · Main Project 1 | Day 125 · Program end

WHAT YOU WILL LEARN

You will learn Python for analytics — from variables and control flow to OOP, files and connecting to a MySQL database.

KEY TOPICS

- Analytics fundamentals
- Variables & data types
- Operators
- Conditions
- Loops
- Functions
- Lambda
- Recursion
- List, tuple
- Dictionary, set
- Random numbers
- Class & object
- Polymorphism
- Abstraction & encapsulation
- File I/O
- Exception handling
- Logical programs
- Patterns
- Connecting Python to MySQL

PRACTICAL OUTCOMES

01 Python exercises

02 OOP & file programs

03 Python–MySQL connection

PHASE 06

06 Python Libraries & EDA

Days 89–103 | **30** HOURS

01 | 02 | 03 | 04 | 05 | **06** | 07

Day 1 — Day 46 · Main Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will use statistics, NumPy, Pandas, Matplotlib and Seaborn to explore real data and present insights.

KEY TOPICS

Statistics for analysts · EDA in Python · NumPy arrays · Indexing · Broadcasting · Series & DataFrames · pipe, apply · loc, iloc, at · Missing values & duplicates · Reading CSV & Excel · Why visualisation matters · Matplotlib · Customising charts · Seaborn · Distribution & relationship plots · Regular expressions · Exploratory data analysis · EDA · Pandas · Visualisation · Real-world dataset · Insight presentation

PRACTICAL OUTCOMES

- **01** Cleaned datasets
- **02** Visualisation set
- **03** EDA mini project
- **04** EDA & insight presentation

PHASE 07

07 AI, Machine Learning & GenAI

Days 104–125 | **44** HOURS

01 | 02 | 03 | 04 | 05 | 06 | **07**

Day 1 — Day 46 · Main Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will build and evaluate core machine learning models with scikit-learn, and learn to use Generative AI for SQL, Python and reporting.

KEY TOPICS

AI introduction | Types of ML | KNN | Evaluation metrics | Naïve Bayes | Support Vector Machines | scikit-learn | Linear regression | Logistic regression | Performance evaluation | Encoding | Ensembles & XGBoost | Decision trees | Random Forest | K-Means | PCA | Role of GenAI in analytics | Prompt engineering for SQL & Python | Hallucination | AI tools | LLM-assisted EDA & reporting | ML model | Real-world dataset | Big data concepts

FINAL OUTCOMES

01 Classification & regression models

02 Clustering & PCA

03 AI-assisted analysis

04 ML + AI main project

DEMONSTRABLE PROJECT WORK

Four main projects

Days 45–46

MAIN PROJECT 1

Multi-Tool Dashboard

YOU DEMONSTRATE

Data cleaning · Data modelling · Interactive dashboards · Storytelling · Excel · Tableau · Power BI · Power Query

Days 69–70

MAIN PROJECT 2

SQL Analysis Project

YOU DEMONSTRATE

Joins & subqueries · Aggregations · SQL to Power BI · Insights · SQL · MySQL · Power BI

Days 102–103

MAIN PROJECT 3

EDA & Insight Presentation

YOU DEMONSTRATE

Data cleaning · EDA · Visualisation · Insight presentation · Python · Pandas · NumPy · Matplotlib

Days 123–124

MAIN PROJECT 4

ML Model with AI Tools

YOU DEMONSTRATE

Feature preparation · Model building · Evaluation · AI-assisted reporting · scikit-learn · ML algorithms · AI tools · LLM-assisted EDA

CHECKING YOUR PROGRESS

Reviews & interview readiness

Your progress is reviewed at the end of every major module, and you will face a mock interview before placement.

CHECKPOINT	WHEN	COVERS
Internal review	**Day 14**	Excel
Internal review	**Day 25**	Tableau
Internal & external review	**Days 47–48**	Excel, Tableau & Power BI
Mock interview	**Day 49**	Interview readiness
Final internal & external review	**Day 122**	SQL, Python, ML & portfolio

REVIEWS

Trainers check your understanding, your code and your project progress — and tell you exactly what to improve.

MOCK INTERVIEW

Practise a real technical and HR-style interview before you apply for jobs.

PLACEMENT SUPPORT AND CAREER ASSISTANCE

Kompetenzen digital career & placement ecosystem

As a student, you can access the Kompetenzen digital placement ecosystem at www.kompetenzen.in for:

01
Job discovery
Students can explore job opportunities listed on the Kompetenzen platform.

02
Direct application
Students can apply directly to suitable job opportunities.

03
ATS resume
Students can create and maintain an ATS-friendly resume.

04
Career profile
Students can maintain their professional profile for placement opportunities.

05
Consultant support
Students can contact Kompetenzen career/placement consultants for guidance and assistance.

06
Interview readiness
Reviews and the mock interview prepare students for applications and interviews.

These are placement support and career assistance services. They do not constitute a guarantee of employment.

YOUR PORTFOLIO PROGRESSION

What you will be able to build

01 EXCEL ANALYSIS WORKBOOK	02 EXCEL MINI PROJECT	03 TABLEAU DASHBOARD
04 POWER BI REPORT WITH RLS	05 MULTI-TOOL DASHBOARD (MAIN PROJECT 1)	06 SQL ANALYSIS (MAIN PROJECT 2)
07 PYTHON DATA PROGRAMS	08 EDA MINI PROJECT	09 EDA INSIGHT PRESENTATION (MAIN PROJECT 3)
10 ML + AI PROJECT (MAIN PROJECT 4)		

FINAL STUDENT OUTCOME

> By the end of the programme, students will have developed the skills to analyse data in Excel, build dashboards in Tableau and Power BI, query databases with SQL, explore and visualise data with Python, build and evaluate machine learning models, use Generative AI responsibly and present insights from real-world data.

Clean. **Analyse.** **Visualise.** **Predict.**